

# MATLAB - Commands

MATLAB is an interactive program for numerical computation and data visualization. You can enter a command by typing it at the MATLAB prompt '>>' on the **Command Window**.

In this section, we will provide lists of commonly used general MATLAB commands.

## Commands for Managing a Session

MATLAB provides various commands for managing a session. The following table provides all such commands –

| Command | Purpose                                   |
|---------|-------------------------------------------|
| clc     | Clears command window.                    |
| clear   | Removes variables from memory.            |
| exist   | Checks for existence of file or variable. |
| global  | Declares variables to be global.          |
| help    | Searches for a help topic.                |
| lookfor | Searches help entries for a keyword.      |
| quit    | Stops MATLAB.                             |
| who     | Lists current variables.                  |
| whos    | Lists current variables (long display).   |

## Commands for Working with the System

MATLAB provides various useful commands for working with the system, like saving the current work in the workspace as a file and loading the file later.

It also provides various commands for other system-related activities like, displaying date, listing files in the directory, displaying current directory, etc.

The following table displays some commonly used system-related commands –

| Command | Purpose                    |
|---------|----------------------------|
| cd      | Changes current directory. |

|         |                                                  |
|---------|--------------------------------------------------|
| date    | Displays current date.                           |
| delete  | Deletes a file.                                  |
| diary   | Switches on/off diary file recording.            |
| dir     | Lists all files in current directory.            |
| load    | Loads workspace variables from a file.           |
| path    | Displays search path.                            |
| pwd     | Displays current directory.                      |
| save    | Saves workspace variables in a file.             |
| type    | Displays contents of a file.                     |
| what    | Lists all MATLAB files in the current directory. |
| wklread | Reads .wk1 spreadsheet file.                     |

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## Input and Output Commands

MATLAB provides the following input and output related commands –

| Command | Purpose                                      |
|---------|----------------------------------------------|
| disp    | Displays contents of an array or string.     |
| fscanf  | Read formatted data from a file.             |
| format  | Controls screen-display format.              |
| fprintf | Performs formatted writes to screen or file. |
| input   | Displays prompts and waits for input.        |
| ;       | Suppresses screen printing.                  |

The **fscanf** and **fprintf** commands behave like C scanf and printf functions. They support the following format codes –

| Format Code | Purpose |
|-------------|---------|
|-------------|---------|

|           |                                                          |
|-----------|----------------------------------------------------------|
| <b>%s</b> | Format as a string.                                      |
| <b>%d</b> | Format as an integer.                                    |
| <b>%f</b> | Format as a floating point value.                        |
| <b>%e</b> | Format as a floating point value in scientific notation. |
| <b>%g</b> | Format in the most compact form: %f or %e.               |
| <b>\n</b> | Insert a new line in the output string.                  |
| <b>\t</b> | Insert a tab in the output string.                       |

The format function has the following forms used for numeric display –

| <b>Format Function</b> | <b>Display up to</b>                 |
|------------------------|--------------------------------------|
| format short           | Four decimal digits (default).       |
| format long            | 16 decimal digits.                   |
| format short e         | Five digits plus exponent.           |
| format long e          | 16 digits plus exponents.            |
| format bank            | Two decimal digits.                  |
| format +               | Positive, negative, or zero.         |
| format rat             | Rational approximation.              |
| format compact         | Suppresses some line feeds.          |
| format loose           | Resets to less compact display mode. |

## Vector, Matrix and Array Commands

The following table shows various commands used for working with arrays, matrices and vectors –

| <b>Command</b> | <b>Purpose</b>                     |
|----------------|------------------------------------|
| cat            | Concatenates arrays.               |
| find           | Finds indices of nonzero elements. |
| length         | Computes number of elements.       |

|          |                                                  |
|----------|--------------------------------------------------|
| linspace | Creates regularly spaced vector.                 |
| logspace | Creates logarithmically spaced vector.           |
| max      | Returns largest element.                         |
| min      | Returns smallest element.                        |
| prod     | Product of each column.                          |
| reshape  | Changes size.                                    |
| size     | Computes array size.                             |
| sort     | Sorts each column.                               |
| sum      | Sums each column.                                |
| eye      | Creates an identity matrix.                      |
| ones     | Creates an array of ones.                        |
| zeros    | Creates an array of zeros.                       |
| cross    | Computes matrix cross products.                  |
| dot      | Computes matrix dot products.                    |
| det      | Computes determinant of an array.                |
| inv      | Computes inverse of a matrix.                    |
| pinv     | Computes pseudoinverse of a matrix.              |
| rank     | Computes rank of a matrix.                       |
| rref     | Computes reduced row echelon form.               |
| cell     | Creates cell array.                              |
| celldisp | Displays cell array.                             |
| cellplot | Displays graphical representation of cell array. |
| num2cell | Converts numeric array to cell array.            |
| deal     | Matches input and output lists.                  |
| iscell   | Identifies cell array.                           |

## Plotting Commands

MATLAB provides numerous commands for plotting graphs. The following table shows some of the commonly used commands for plotting –

| Command   | Purpose                                       |
|-----------|-----------------------------------------------|
| axis      | Sets axis limits.                             |
| fplot     | Intelligent plotting of functions.            |
| grid      | Displays gridlines.                           |
| plot      | Generates xy plot.                            |
| print     | Prints plot or saves plot to a file.          |
| title     | Puts text at top of plot.                     |
| xlabel    | Adds text label to x-axis.                    |
| ylabel    | Adds text label to y-axis.                    |
| axes      | Creates axes objects.                         |
| close     | Closes the current plot.                      |
| close all | Closes all plots.                             |
| figure    | Opens a new figure window.                    |
| gtext     | Enables label placement by mouse.             |
| hold      | Freezes current plot.                         |
| legend    | Legend placement by mouse.                    |
| refresh   | Redraws current figure window.                |
| set       | Specifies properties of objects such as axes. |
| subplot   | Creates plots in subwindows.                  |
| text      | Places string in figure.                      |
| bar       | Creates bar chart.                            |
| loglog    | Creates log-log plot.                         |
| polar     | Creates polar plot.                           |

|          |                                               |
|----------|-----------------------------------------------|
| semilogx | Creates semilog plot. (logarithmic abscissa). |
| semilogy | Creates semilog plot. (logarithmic ordinate). |
| stairs   | Creates stairs plot.                          |
| stem     | Creates stem plot.                            |